HTTP vs HTTPS

HTTP vs HTTPS — Core Difference

Feature	HTTP	HTTPS
Full form	**HyperText Transfer Protocol**	**HyperText Transfer Protocol Secure**
Security	No encryption	Encrypted communication
Data transfer	Plain text	Encrypted using SSL/TLS
Port number	80	443
URL format	http://example.com	https://example.com
Certificate	Not required	Requires SSL/TLS certificate
Data safety	Data can be intercepted	Data is protected

HTTP:

HTTP is the **basic protocol used to transfer web pages between a browser and a server**.

How it works

1. Browser sends a request
2. Server responds with web content

Example request:

```
GET /index.html HTTP/1.1
Host: example.com
```

Problem

Data is sent in **plain text**, so attackers can:

- Read the data
- Modify the data
- Steal login credentials

HTTPS:

HTTPS is **HTTP with encryption using SSL/TLS**.

It uses **TLS** (earlier **SSL**).

What HTTPS provides

1. **Encryption** – Data cannot be read by attackers
2. **Authentication** – Server identity verified using certificates
3. **Integrity** – Data cannot be modified during transmission

Example

https://google.com

Browser shows a **lock icon** indicating secure communication.

Simple Real-World Analogy

- **HTTP** → Sending a postcard (anyone can read it)
- **HTTPS** → Sending a sealed envelope (only receiver can read it)